

QXU4006 Materials Science 2: Coursework Assessment Brief (40%)

1. Overview – Updated Coursework Structure

This coursework involves modifying a standard ASTM D638 tensile test specimen using SolidWorks, followed by a comparative engineering study to evaluate the mechanical behavior of Acrylonitrile Butadiene Styrene (ABS) and Polylactic Acid (PLA). As the 3D printers are currently unavailable, you will not be required to physically print or test your design. Instead, your assessment will be based on your CAD modifications, analysis of material properties, and engineering reasoning. This integrated approach allows you to demonstrate your understanding of material behavior and design application without needing physical fabrication.

The coursework consists of the following components:

- Modify a 3D design using SolidWorks (available through Queen Mary).
- Compare ABS and PLA based on their material properties and how they affect your design.
- Submit a written engineering report and a group reflection video.
- Complete a peer review assessment on QMPlus.

Deadline: 30 May at 23:59 (China Time)

Submission: Submit via QMPlus

One Submission per Group: Only the group representative should upload the submission.

Similarity Check: Your work will be automatically checked for similarity on QMPlus. The similarity index should be below 20%. Plagiarism, including content generated by GenAI, is strictly prohibited.

Signatures Required: All group members must include their names and signatures on the report before submission.

2. Task Breakdown

Your coursework is divided into the following key tasks:

A. Modify the ASTM D638 Standard Tensile Test Specimen:

- i. Use the ASTM D638 tensile test specimen uploaded on QMPlus as a baseline and apply modifications to suit a specific real-world application.
- ii. Examples of modifications to suit practical applications:
 - Aerospace Application: For example, extend the grip areas to enhance load distribution in high-stress environments.
 - Medical Application: For example, adjust the thickness to replicate orthopedic supports for improved structural integrity.
- iii. All modifications must be justified in the report based on engineering principles.
- iv. Include screenshots of your modified model in the report.

B – Experimental Dataset Analysis

Use the data provided in Table 1 to analyze how ABS and PLA behave under tensile (pulling) forces.

- i. Estimate and sketch a stress-strain graph using the values from the table.
- ii. Look closely at the following features in both the table and your graph:
 - Tensile strength
 - Elongation at break
 - Stiffness
 - Failure mode
- iii. Based on your observations, answer these questions:
 - Which material is stronger, stiffer, or more brittle?
 - What are the key differences in how ABS and PLA fail?
 - What do these results suggest about which material is better for different real-world applications?

Table 1 – Experimental Data

Material	Max Force (N)	Ultimate Tensile Strength (MPa)	Elongation at Break (%)
ABS	400	35	15
PLA	450	50	5

C – Comparative Engineering Study Using Your Modified Design

In this section, you will analyze how ABS and PLA would perform in the part you have designed.

What You Need to Do:

- i. **Describe your modified design**
 - Explain the changes you made to the original ASTM D638 specimen.
 - State what application your design is intended for (e.g., a sports component, medical support, or aerospace part).
- ii. **Research material properties**

Find reliable sources (e.g., material datasheets, engineering textbooks, etc) to fill in Table 2 with the following properties for ABS and PLA:

 - Elastic Modulus (GPa)
 - Tensile Strength (MPa)
 - Elongation at Break (%)
 - Density (g/cm³)
 - Glass Transition Temperature (°C)
- iii. **Compare ABS and PLA in your design**

Use the values from your table and apply engineering reasoning to answer the following:

 - Which material is better for your design and why?
 - How do their properties (like stiffness, strength, and brittleness) affect the part's performance?

- Would the part fail differently depending on the material? What would that failure look like?
- Are there parts of your design that are more likely to fail (e.g., thinner sections or hollow areas)?

Table 2 – Material Properties (Fill this in with your research)

Property	ABS	PLA
Elastic Modulus (GPa)		
Tensile Strength (MPa)		
Elongation at Break (%)		
Density (g/cm ³)		
Glass Transition Temp (°C)		

3. Submission Guidelines: Report and Group Video

Model and Engineering Report – 50% of Coursework

- **Modified design:** 3D design modified SolidWorks model - **(10%)**
- **Engineering Report:** 3 Pages (PDF Format) - **(40%)**

Your written report (3 pages) should be well-organized and include:

Page 1:

- Introduction and design objectives
- Description of your modifications
- Clear screenshots of your SolidWorks model

Page 2:

- Explanation of the real-world function your design is intended for
- A table comparing material properties of ABS and PLA
- Engineering analysis of how each material would perform in your design

Page 3:

- Justify which material is more suitable and why
- References for any data or material sources used (e.g., datasheets, textbooks)

File naming format:

- **Materials Group:** QXU4006_B24_M1 (replace “M1” with your actual group number)
- **Polymers Group:** QXU4006_B24_P7 (replace “P7” with your actual group number)

Important: Each group member must write their name and **sign the report** before the group representative submits it via QMPlus.

Group Reflection Video – 4 to 5 Minutes (MP4 Format) – 37.5% of Coursework

Your group must create a video where each member participates equally. Keep it engaging, clear, and well-paced.

Requirements:

- Each group member must speak for approximately 1 minute
- Show the names of all group members clearly on the screen
- Save the video using the same naming format as the report

What to include in the video:

- Walk through your design process and the reasons behind your modifications
- Reflect on what you learned about PLA vs. ABS and how it influenced your choices
- Mention any challenges, surprises, or group insights along the way

The video should feel like a shared story – explain your design journey from start to finish.

4. Peer Review Assessment – 12.5% of Coursework

- Each student must complete a peer evaluation form to assess the contributions of their other group members.
- The link to the form will be provided on QMPlus.
- You'll rate your teammates based on their effort, participation, and teamwork.

This is an important part of your assessment, so please complete it thoughtfully and honestly.

Final Submission Checklist

Before submitting, make sure you have completed everything below:

- i. Your report includes:
 - Screenshots of your SolidWorks model
 - Material property tables
 - Engineering analysis and discussion
- ii. The report is signed by all group members and clearly shows group names.
- iii. Your video is complete, correctly named, and each member appears and speaks.
- iv. You have submitted your peer review form on QMPlus.

Innovate, experiment, and excel in your designs!